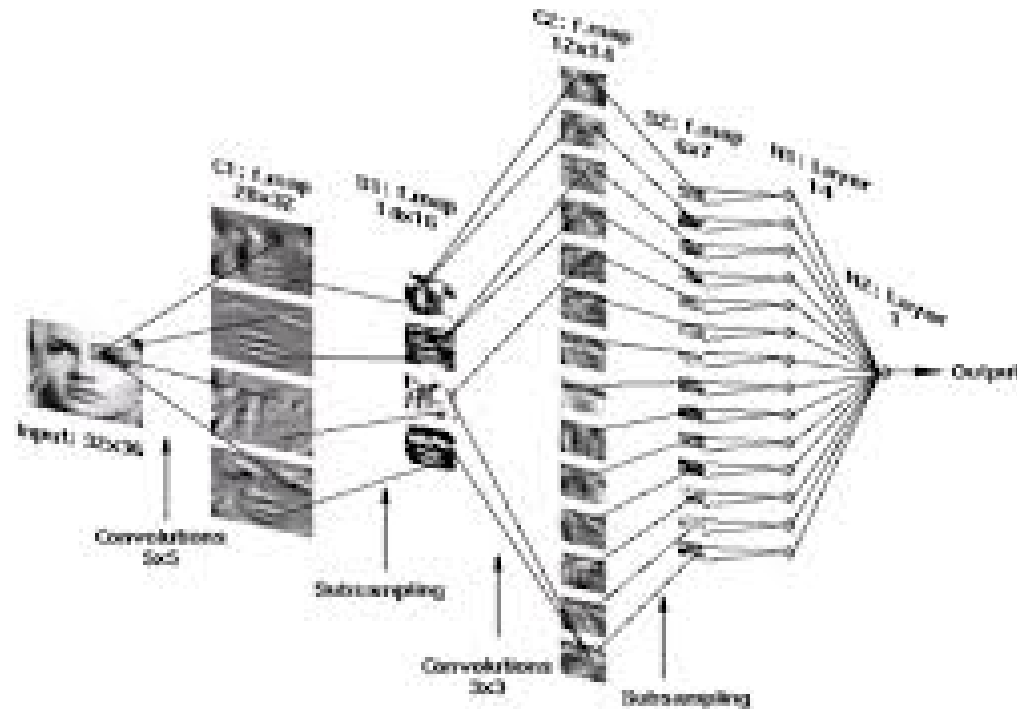


Mathematics of Machine Learning

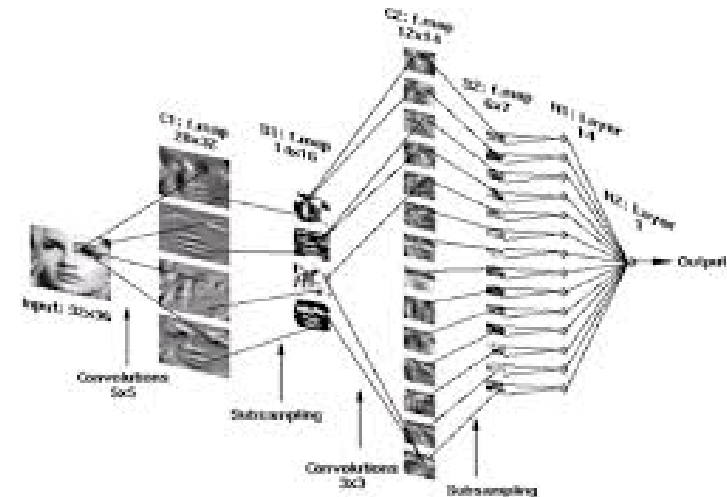
Introduction



Course Details and Topics

Outline

1. **Why are you doing this course?**
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation
- Notes are on Moodle and

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- LLMs can help us a lot, but if you don't trust yourself you are disempowered
- This course is aimed at building up trust in your understanding of how machine learning works

Why Mathematics?

- Learning machines are notoriously difficult to understand
- Mathematics allows us to reason rigorously
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition
- Mathematics also allow us to describe the world precisely
- It also provides an arsenal of tools that turn out to be hugely useful

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Cracking the Code

- Mathematics is a language that takes time to master
- For some it is easy, for others it will be a struggle
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Starting from Different Places

- You all start from different places
- Some of you may have a degree in mathematics
- For others it may have been your least favourite subject
- I invite strong mathematicians to become even stronger
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others

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The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides
- The journey will pay you off, these are the foundations of most of machine learning and will give you super-powers that are hard to acquire without mathematics

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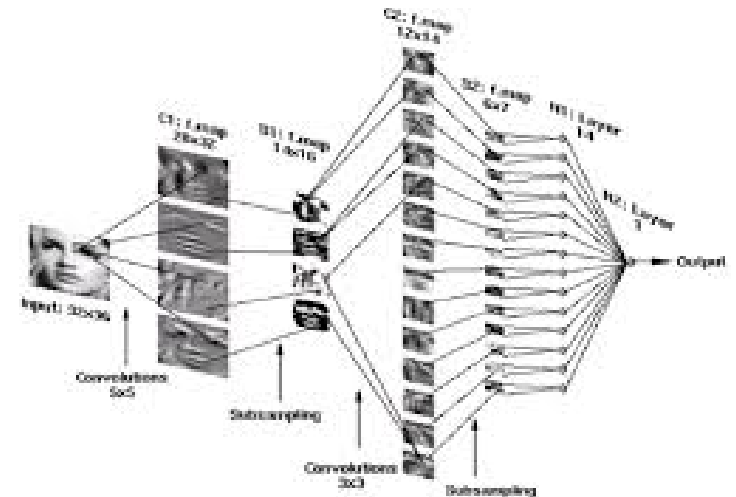
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What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni
- You should know what a function is, e.g. $f(x) = x^2 + 3$
- You should be able to do simple algebraic manipulations
- And some elementary calculus

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Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar
- We will use bold symbols to represent **column vectors**, e.g.

$$\mathbf{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- We denote **row vectors** as the **transpose** of column vectors
 $\mathbf{x}^T = (1.2, 2.3, 4.7)$
- We care whether vectors are row or column vectors as it allows us to write equations involving vectors succinctly and unambiguously

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Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
- An $m \times n$ matrix acting on a vector in $\mathbb{R}^n$ transforms it to $\mathbb{R}^m$ vector, e.g.

$$\mathbf{M}x = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix} \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix} = \begin{pmatrix} 28.63 \\ -2.39 \end{pmatrix}$$

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Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$
 - ★ and then applying $\mathbf{A}$ to get vectors in $\mathbb{R}^\ell$

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Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

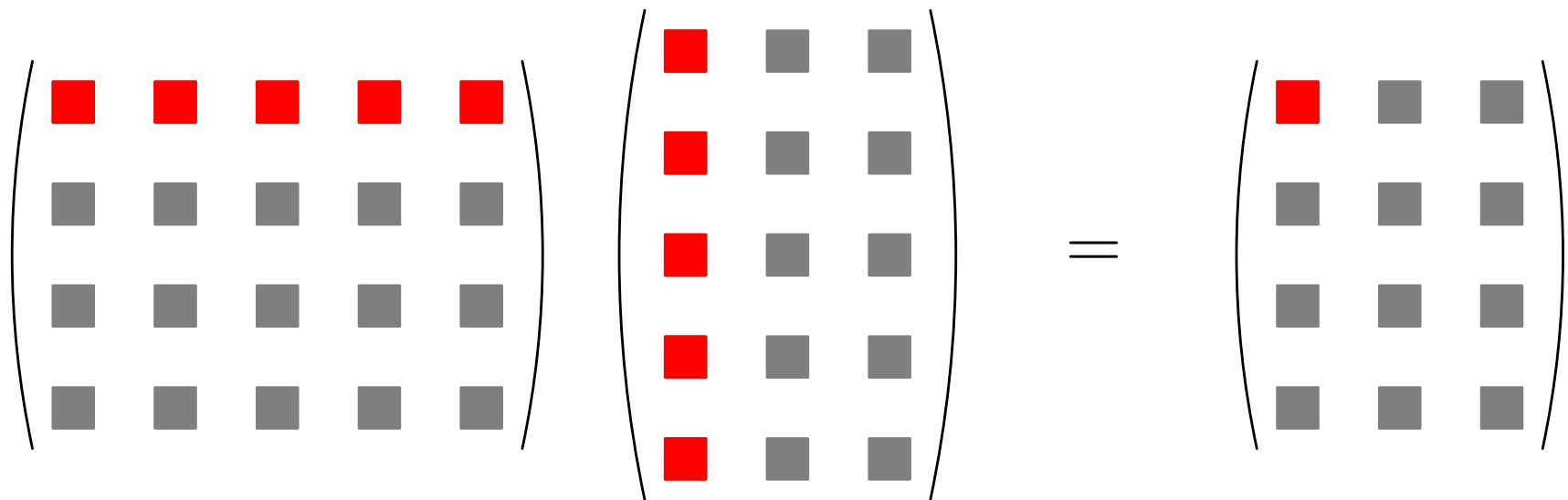
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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. On the left, a 5x5 matrix (A) is shown with 5 rows and 5 columns of squares. This is followed by a 5x3 matrix (B) with 5 rows and 3 columns of squares. An equals sign follows, leading to a 5x3 matrix (C) with 5 rows and 3 columns of squares. The visual representation shows that the product of a 5x5 matrix and a 5x3 matrix is a 5x3 matrix.

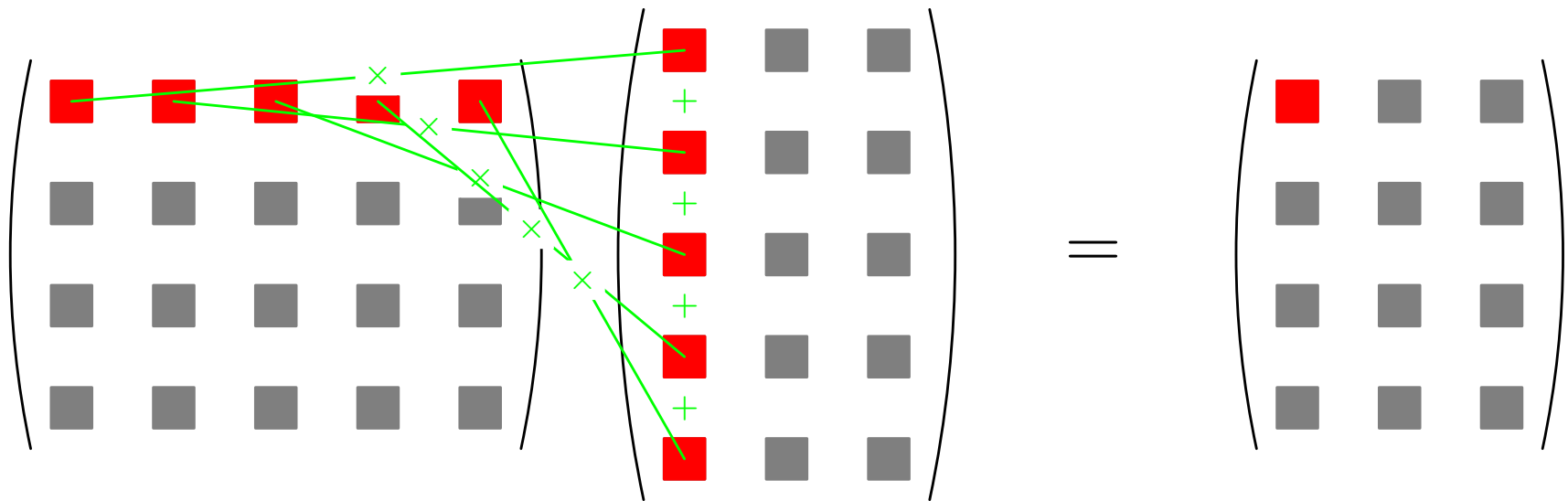
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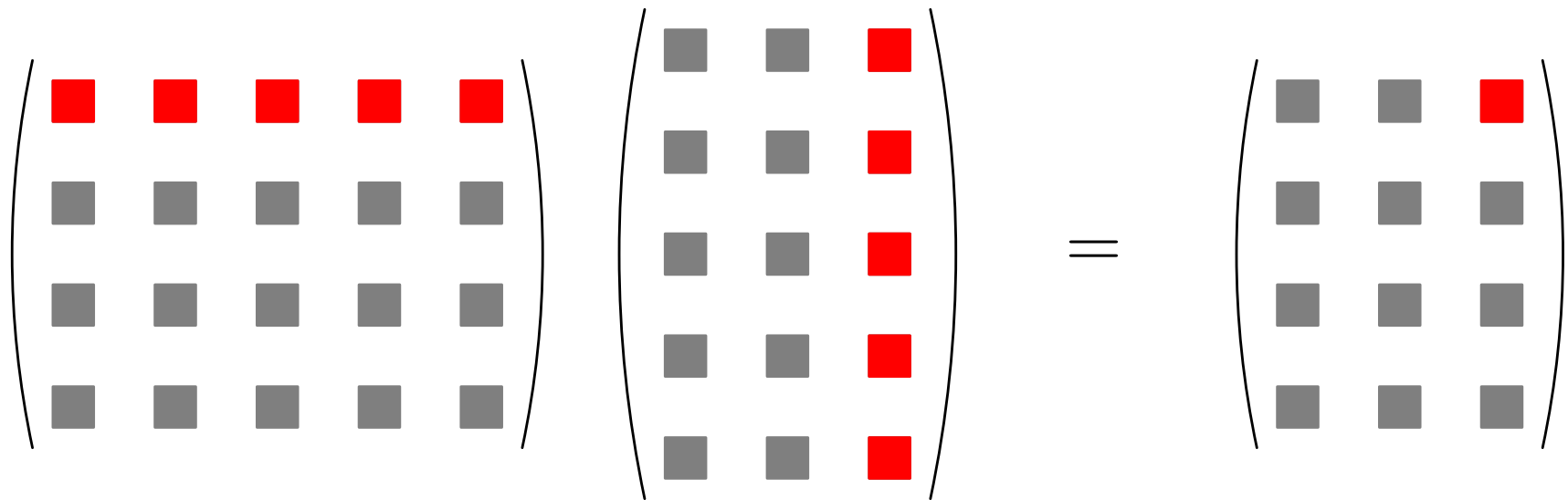
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The diagram illustrates the matrix multiplication $\mathbf{AB} = \mathbf{C}$ using a grid of squares. Matrix $\mathbf{A}$ is a 4×5 matrix with 5 red squares in the first row and 4 gray squares in each of the other three rows. Matrix $\mathbf{B}$ is a 5×3 matrix with 5 red squares in the second column and 2 gray squares in each of the other two columns. Matrix $\mathbf{C}$ is a 4×3 matrix with 1 red square in the second row, second column and 4 gray squares in the other three rows, second column. The matrices are enclosed in large parentheses, and an equals sign is placed between $\mathbf{B}$ and $\mathbf{C}$.

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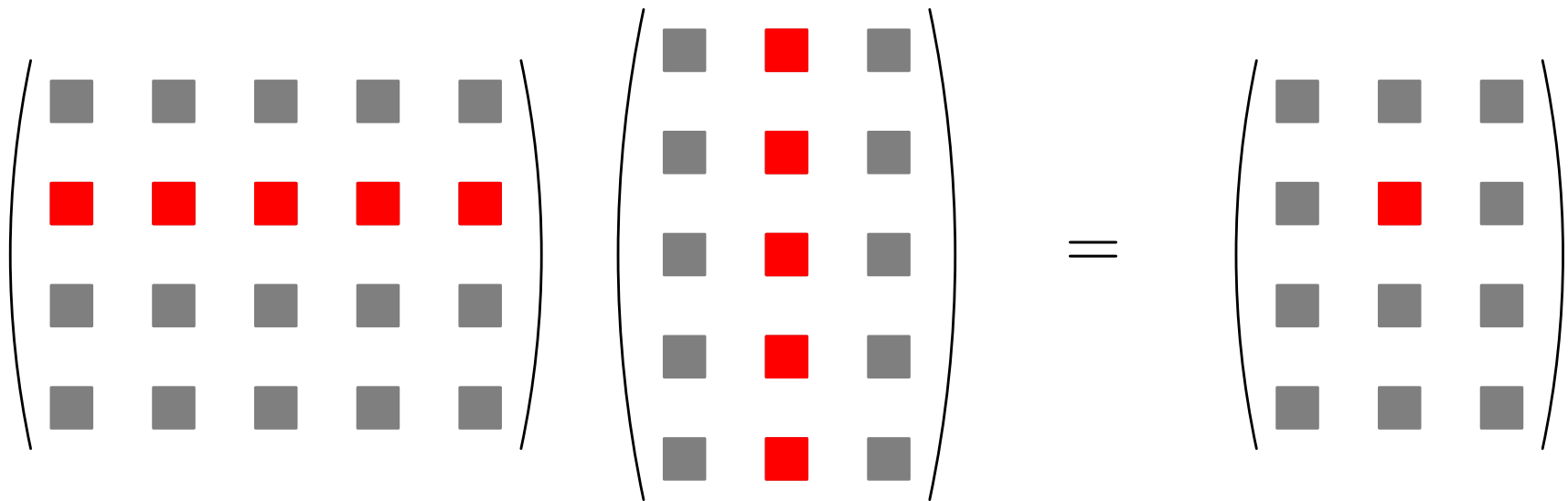
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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with gray squares, where the second row, first column square is red. The second matrix is a 5x3 grid with gray squares, where the first, second, third, fourth, and fifth rows, first column squares are red. An equals sign follows, and then a 4x3 grid with gray squares, where the second row, first column square is red.

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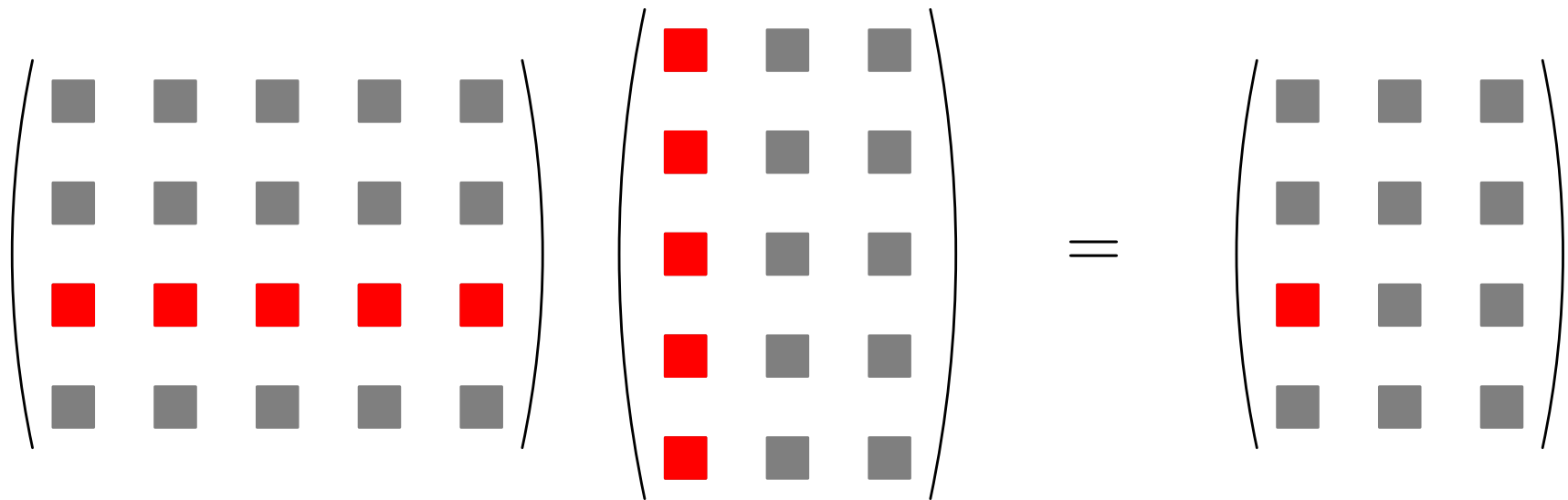
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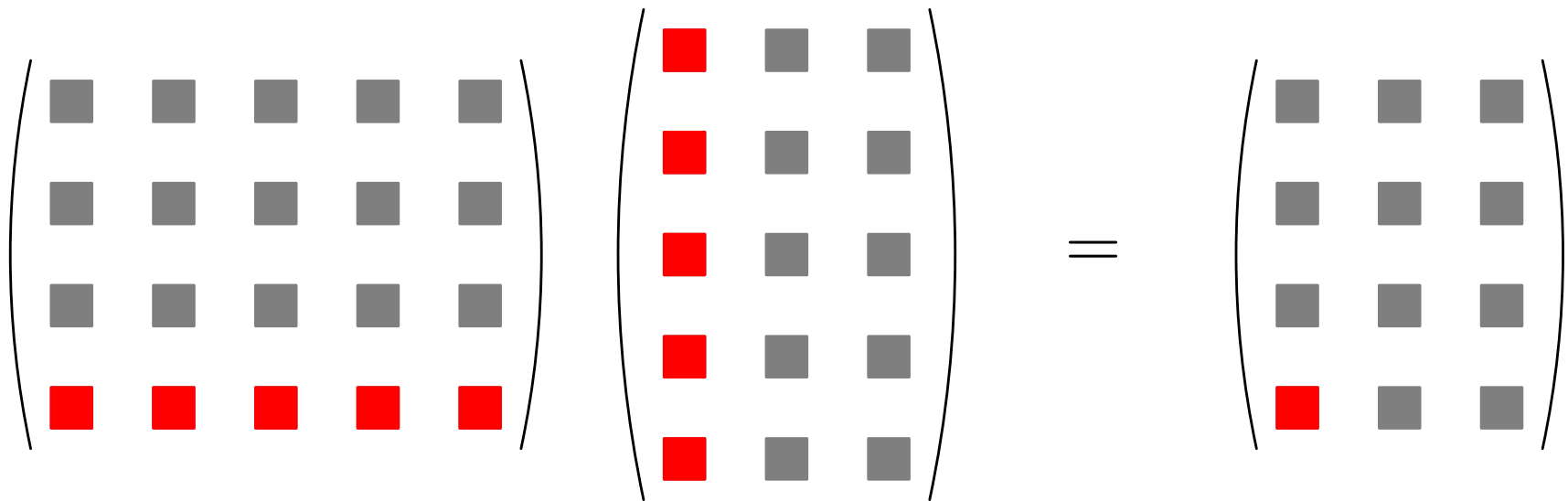
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The diagram illustrates the matrix multiplication $\mathbf{AB} = \mathbf{C}$ using a grid of squares. Matrix $\mathbf{A}$ is a 4×5 matrix with 4 rows and 5 columns. Matrix $\mathbf{B}$ is a 5×3 matrix with 5 rows and 3 columns. Matrix $\mathbf{C}$ is a 4×3 matrix with 4 rows and 3 columns. The squares are colored gray or red. In $\mathbf{A}$, the third row is red. In $\mathbf{B}$, the third column is red. In $\mathbf{C}$, the third column is red. This represents the calculation of the third column of $\mathbf{C}$ as the sum of the products of the rows of $\mathbf{A}$ and the third column of $\mathbf{B}$.

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$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \end{pmatrix}$$

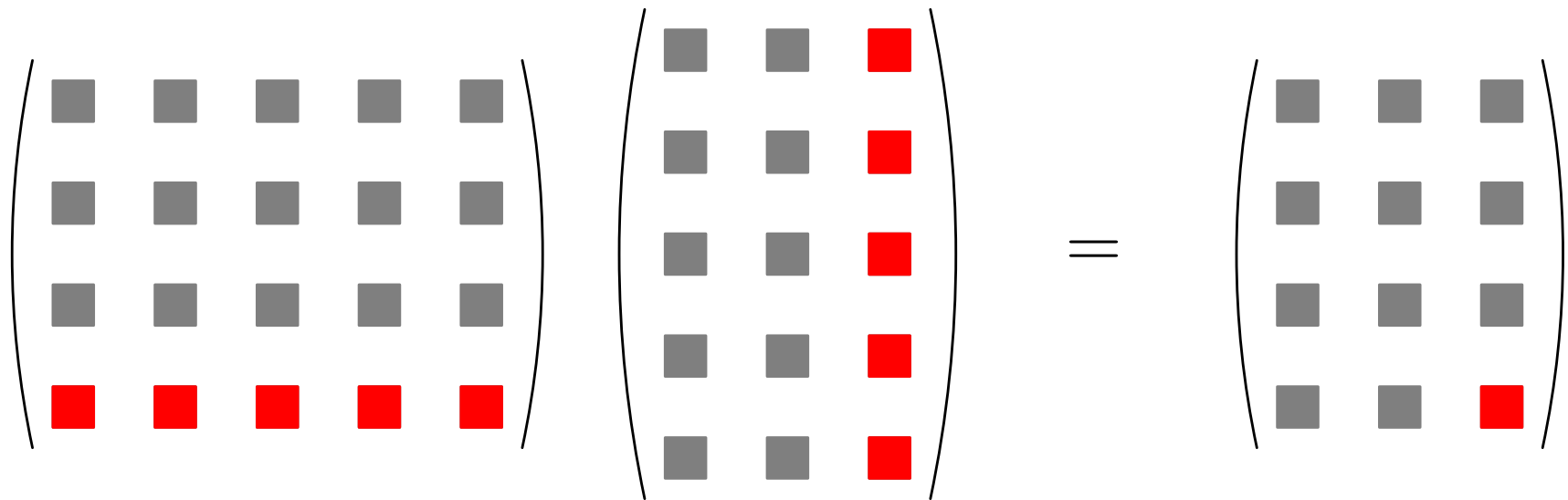
Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The first three rows are all gray squares, and the fourth row consists of five red squares. The second matrix is a 5x3 grid with 5 rows and 3 columns. The first four rows consist of two gray squares followed by one red square, and the fifth row consists of two gray squares followed by one red square. An equals sign follows, and the resulting matrix is a 4x3 grid with 4 rows and 3 columns. The first three rows consist of three gray squares, and the fourth row consists of two gray squares followed by one red square.

Matrix Multiplication

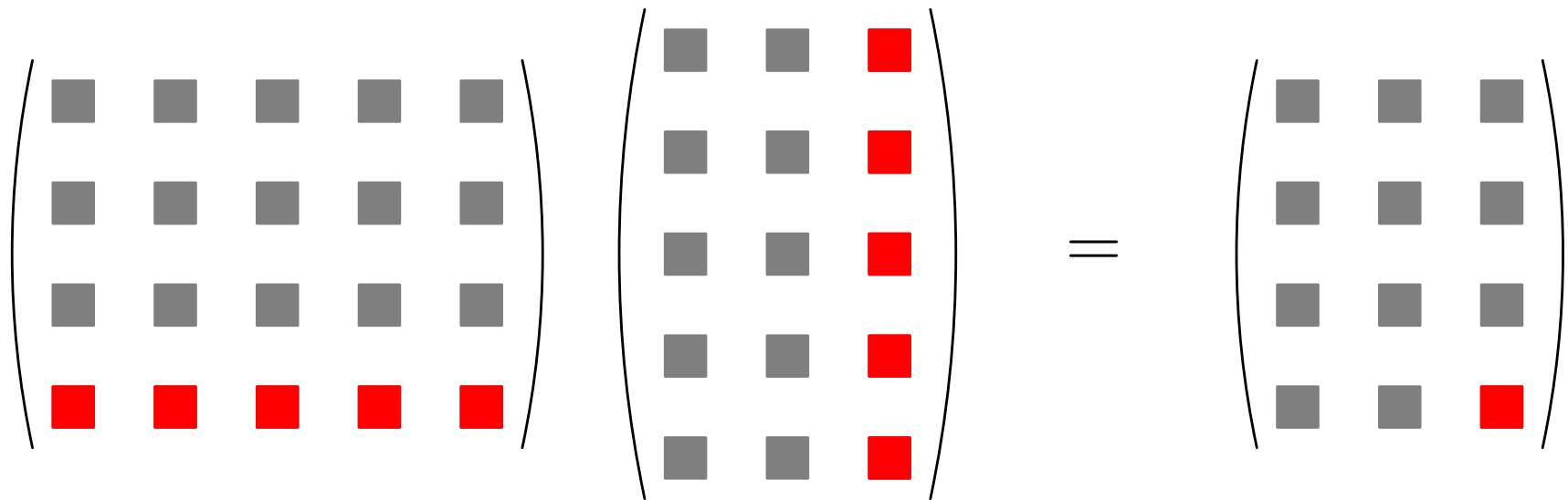
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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

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- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)



- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\mathbf{a}^T \mathbf{b} = (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a}$$

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- We often encounter transposes of a matrix $\mathbf{M}^\top$ where we have exchanged the rows and columns: $[\mathbf{M}^\top]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^\top)^\top = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^\top$
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Transpose

- We often encounter transposes of a matrix $\mathbf{M}^\top$ where we have exchanged the rows and columns: $[\mathbf{M}^\top]_{ij} = [\mathbf{M}]_{ji}$
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Transpose

- We often encounter transposes of a matrix $\mathbf{M}^\top$ where we have exchanged the rows and columns: $[\mathbf{M}^\top]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^\top)^\top = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^\top$
- Since 1×1 matrices are always symmetric

$$\begin{aligned} \mathbf{a}^\top \mathbf{b} &= (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{M} \mathbf{b} &= \mathbf{b}^\top \mathbf{M}^\top \mathbf{a} \end{aligned}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is represented by a grid of 20 gray squares enclosed in large parentheses. To its right is a 5x1 column vector, represented by a single column of 5 gray squares enclosed in large parentheses. An equals sign follows, and to the right is the resulting 4x1 column vector, represented by a single column of 4 gray squares enclosed in large parentheses.

Matrix-Vector Multiplication

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- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 4x5 matrix (with 5 red squares in the top row and 4 gray squares in each of the other three rows) multiplied by a 5x1 column vector (with 5 red squares). This is followed by an equals sign and a 4x1 column vector (with 1 red square in the top row and 3 gray squares in the other rows), representing the result of the multiplication.

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of squares, with the second row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of squares, with all squares highlighted in red). The result is a 5x1 column vector (represented by a single column of squares, with the second square highlighted in red). The equation is represented by an equals sign between the two sides.

Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 5x5 matrix is shown with 4 gray squares and 1 red square in each row. This matrix is multiplied by a 5x1 column vector consisting of 5 red squares. The result is a 5x1 column vector consisting of 4 gray squares and 1 red square. The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of 25 squares, with the bottom row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). The result is a 5x1 column vector (represented by a single column of 5 squares, with the bottom square highlighted in red). The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{5x5 Matrix} \end{pmatrix} \begin{pmatrix} \text{5x1 Column Vector} \end{pmatrix} = \begin{pmatrix} \text{5x1 Column Vector} \end{pmatrix}$$

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $a^T b$

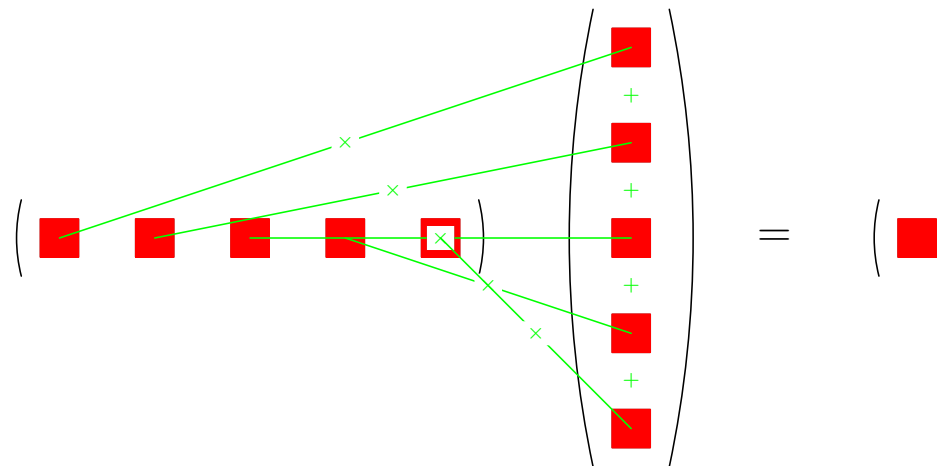
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$$\left(\begin{array}{ccccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{array} \right) = \left(\begin{array}{c} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{array} \right) = \left(\blacksquare \right)$$

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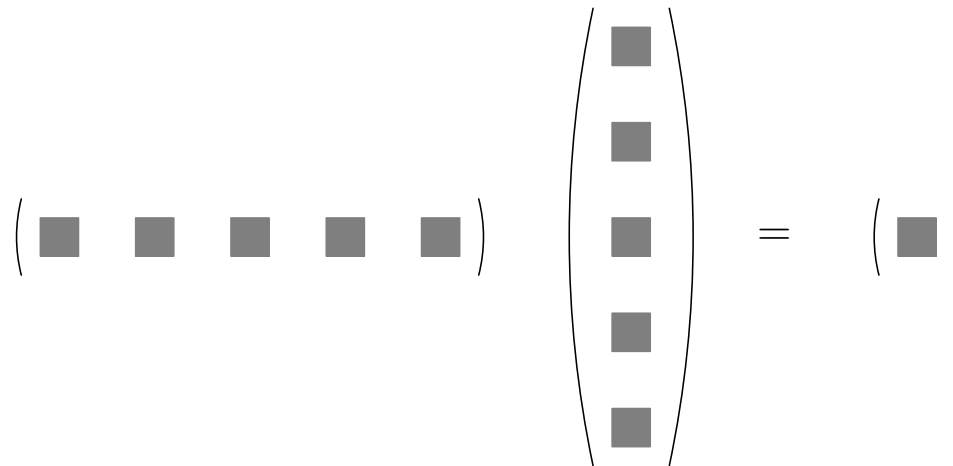
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- Inner products encode angles between vectors

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is represented by a large left parenthesis followed by four gray squares stacked vertically, and a large right parenthesis. This is multiplied by a row vector b^T of size 1x3, represented by a large left parenthesis followed by three gray squares, and a large right parenthesis. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is shown as a large left parenthesis followed by four rows, each containing three gray squares, and a large right parenthesis.

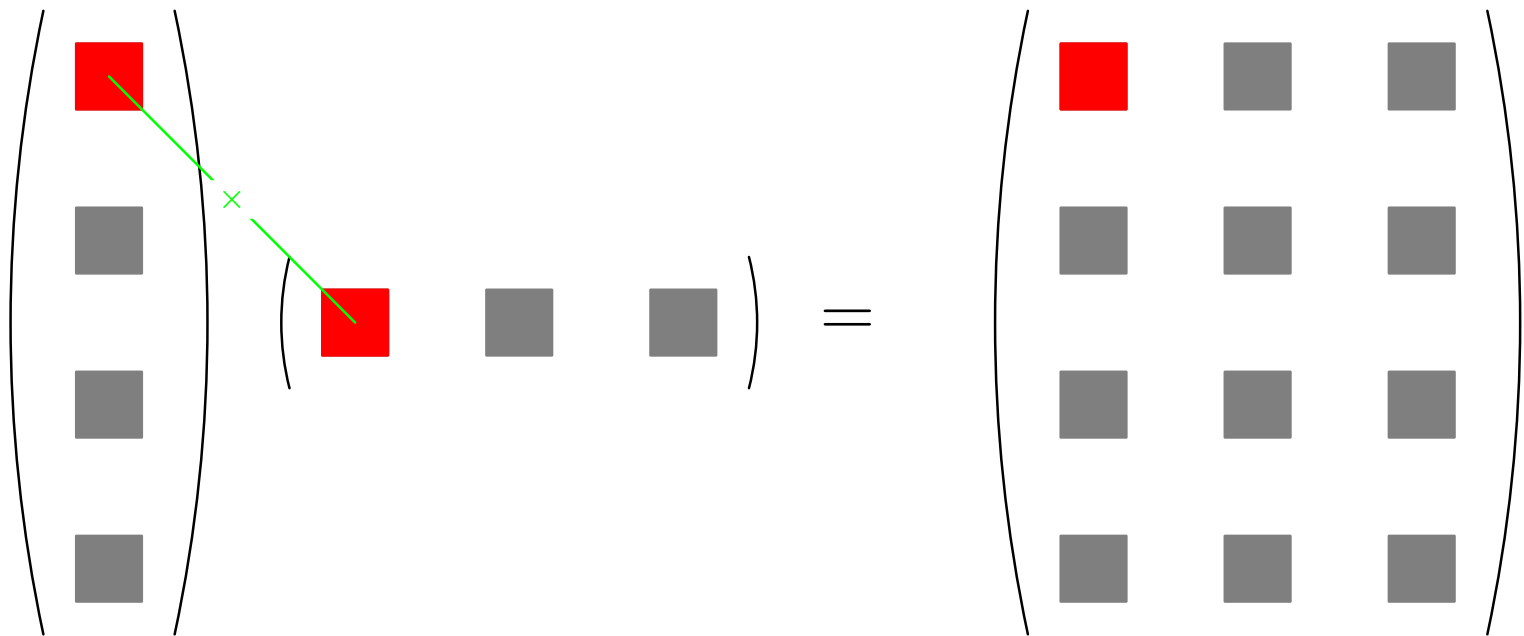
Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector b^T of size 1x3, also with its first element highlighted in red. The result is a 4x3 matrix where each row is a scalar multiple of the vector b^T . The first row of the resulting matrix has its first element highlighted in red, while the other rows have their first elements in gray, indicating they are scaled versions of the first row.

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector of size 1x3, also with its middle element highlighted in red. The result is a 4x3 matrix where every element in the first column is red, representing the scalar product of the original red elements multiplied by each row of the second vector.

$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product of two vectors. On the left, a column vector is shown with four elements: a red square at the top, followed by three gray squares. This is multiplied by a row vector with three elements: two gray squares followed by a red square. The result is a 4x3 matrix on the right, where each row is a scaled version of the row vector. The first row has a red square in the third column, while the other three rows have gray squares in all three columns.

$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T using square blocks to represent elements of matrices and vectors. On the left, a column vector a is shown as a vertical stack of four squares, with the second square from the top highlighted in red. To its right is a row vector b^T shown as a horizontal row of three squares, with the first square from the left highlighted in red. An equals sign follows, leading to a 4x3 matrix on the right. This matrix is formed by scaling the row vector b^T by each element of the column vector a . The first column of the resulting matrix consists of three gray squares, and the second and third columns each consist of one red square followed by two gray squares, mirroring the structure of the original vectors.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its second element highlighted in red. This is multiplied by a row vector of size 3, with its second element also highlighted in red. The result is a 4x3 matrix where every element in the second row and second column is highlighted in red, representing the outer product of the two vectors.

$$\begin{pmatrix} \square \\ \color{red}\square \\ \square \\ \square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \color{red}\square & \square \\ \square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a is shown with four elements: three gray squares and one red square in the second position. This is multiplied by a row vector b^T with three elements: two gray squares and one red square in the third position. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from a and b^T . The entire matrix has a red square in the second row and third column, with all other elements being gray squares.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T using square blocks to represent matrix elements. On the left, a column vector a of size 4x1 is shown, with its third element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scalar multiple of the vector b^T , and the third row is highlighted in red because it corresponds to the red element in a .

$$\begin{pmatrix} \square \\ \square \\ \color{red}\square \\ \square \end{pmatrix} \begin{pmatrix} \color{red}\square & \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4x1 is shown, containing four gray squares, with the third square highlighted in red. This is multiplied by a row vector of size 1x3, also containing three gray squares, with the middle square highlighted in red. An equals sign follows, leading to a 4x3 matrix on the right. This matrix is the result of the outer product, where each element is the product of the corresponding elements from the two input vectors. The entire 4x3 matrix is shown within large parentheses, and the red square in the middle row, second column, represents the product of the red squares from the input vectors.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T . On the left, a column vector a is represented by a large left parenthesis followed by four gray squares, with the third square highlighted in red, and a large right parenthesis. This is multiplied by a row vector b^T , represented by a large left parenthesis followed by three squares (two gray, one red), and a large right parenthesis. An equals sign follows. On the right, the resulting 4x3 matrix is shown within large parentheses, consisting of four rows and three columns of squares. The third row and third column are highlighted in red, representing the outer product of the red elements from the original vectors.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T using square blocks to represent matrix elements. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix on the right, where each row is a scaled version of the row vector b^T . The first row of the resulting matrix has its first element highlighted in red, corresponding to the product of the red elements in the input vectors.

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The diagram illustrates the outer product of two vectors. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its middle element highlighted in red. The result is a 4x3 matrix on the right, where the bottom-middle element is red, indicating it is the product of the red elements from the input vectors. All other elements in the matrix are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \color{red}\square & \square \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, consisting of three gray squares and one red square at the bottom. This is multiplied by a row vector of size 3, consisting of two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the two input vectors. The bottom-right element is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product of a column vector and a row vector. On the left, a column vector is shown with four elements: three gray squares and one red square at the bottom. This is multiplied by a row vector with three elements: two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the input vectors. The bottom-right element is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

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Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij} a_j$$

- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

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$$b_i = \sum_j M_{ij}a_j = M_{i1}a_1 + M_{i2}a_2 + \cdots + M_{in}a_n$$

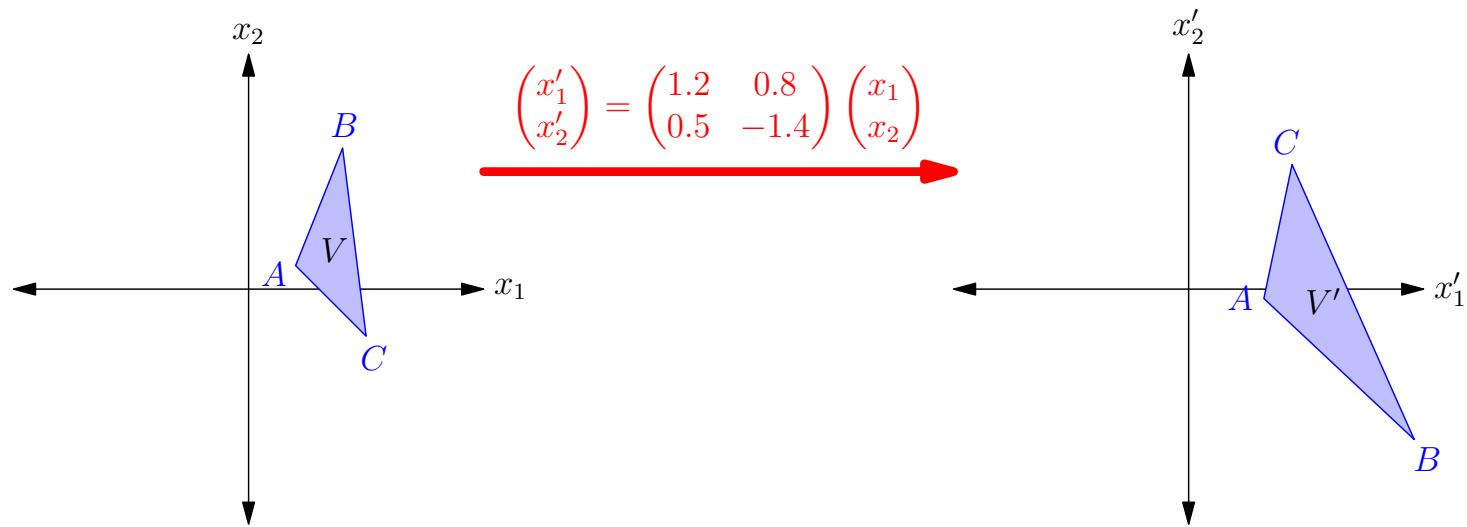
- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

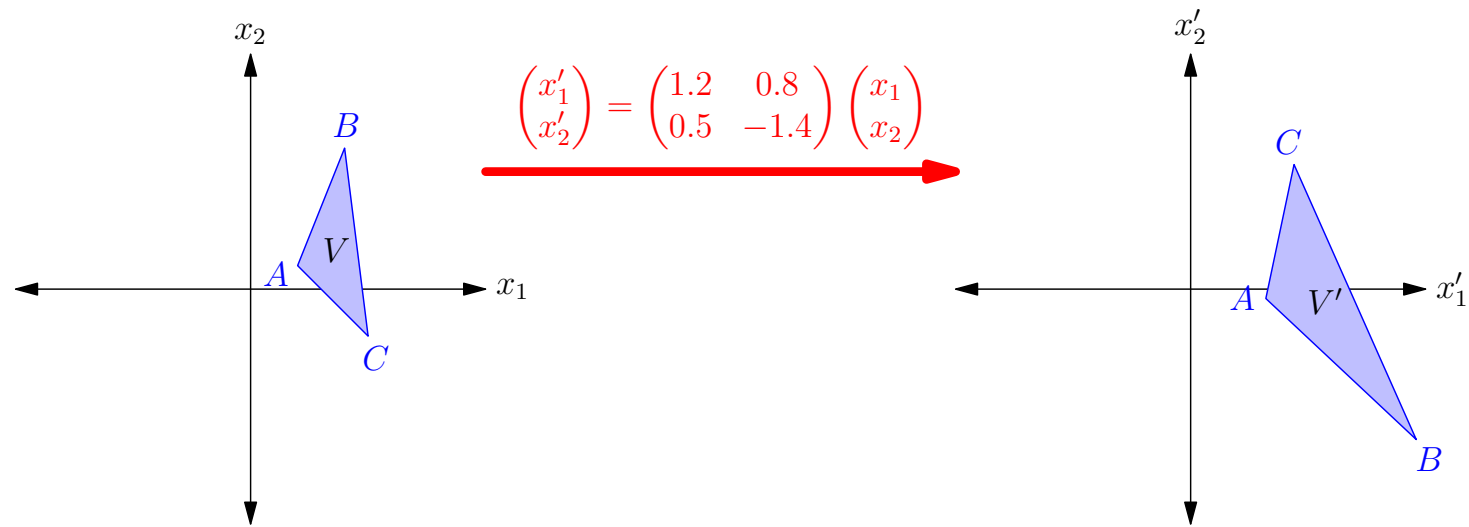
- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

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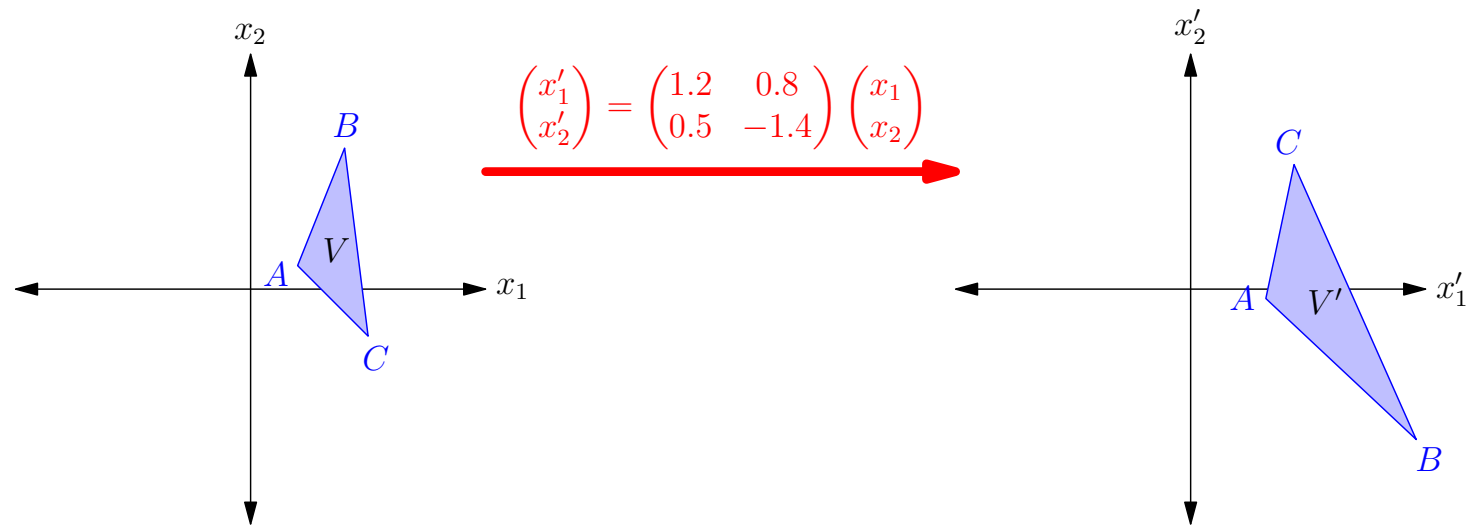
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Determinants

- A consequence of the fact that $\mathbf{AB}$ denotes the composition, apply mapping $\mathbf{B}$ then apply mapping $\mathbf{A}$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$$

- Determinants are only defined for square matrices, as non-square matrices project into different size spaces, where the volumes cannot be compared
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Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- E.g. a learning machine for regression might be $\hat{f}(\mathbf{x}, \mathbf{w})$
- We say a function f is linear if

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Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

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- To reason about events with uncertain outcomes we introduce **random variables**
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event
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- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$
- For continuous events the probability of X taking a particular value x is zero
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$

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Expectations

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$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) g(x) \\ \int f_X(x) g(x) dx \end{cases}$$

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- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered
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